MUHAMMAD MUIZZ ZAFAR

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Software Engineer, Backend | Python | APIs at Scale | AWS | Automation | SQL

Backend software engineer designing, developing and launching Python APIs and backend systems at scale: REST API development, data pipelines, and workflow automation with orchestration and monitoring. SQL-first, with data modeling and reliability practices proven on 20M+ record production systems.

Skills

Languages:	Python, SQL, GoLang, Java, C#, C++
Data Engineering:	Backend Development, REST APIs, Apache Airflow, PySpark, Apache Spark, Data Modeling, API Design AWS (Lambda, S3, Athena, Glue, CloudFormation), MSSQL Server, Oracle, PostgreSQL, Docker, Git, Li

Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - Dec 2024

Work experience

Data Engineer — May 2024 - Present
Government Of Canada — Ottawa, ON

- Designed logical and physical data models, defining schema and relationships for a centralized **Microsoft SQL Server** data warehouse supporting datasets of **20M+ records**.
- Built and maintained scalable **ETL pipelines** to extract data from **Oracle** databases, transform it using **SQL** and **Python**, and load it into semi-structured data in a JSON-based **data lake** layer and Microsoft SQL Server data warehouse, processing **5GB+** of data per pipeline run.
- Developed a reusable ETL validation framework using **Python** and **PySpark**, enabling automated validation of large datasets (**20M+ records**) across dev, uat, and prod environments, reducing data discrepancies by **60%**.
- Implemented data quality checks, scoring logic, and anomaly detection workflows to proactively identify inconsistencies, improving data reliability and reducing production data issues by **50%**.
- Orchestrated and scheduled data workflows using **Apache Airflow**, automating pipeline execution and dependency management, reducing manual intervention by **70%** and improving pipeline reliability.
- Supported data migration projects, validating data quality and integrity during migration and improving consistency across systems.
- Developed and optimized complex **SQL queries and transformations**, reducing processing time for large-scale ETL jobs by **35 minutes**.
- Built **Power BI and Streamlit dashboards** to visualize pipeline performance, data quality metrics, and operational insights, enabling stakeholders to identify issues **2x faster** and support data-driven decision-making.
- Designed, developed and launched production **Python** APIs with **FastAPI** that external client applications integrate with to access electoral address data for Elections Canada's Electronic List of Electors (E-list) project, powering the poll-site electronic voters list.
- Built **Python** backend services and integrations moving data from **Oracle** transactional sources through a JSON data lake into a **SQL Server** warehouse at **5GB+** per run, with schemas and models designed for downstream APIs and analytics.
- Built an end-to-end **Python** automation process that pulls employee data from the Public Service Commission into Elections Canada databases across multiple systems, replacing manual transfers with a reliable automated pipeline.
- Provided on-call support for production data systems and reviewed pull requests and data models for other data architects on the team, supporting reliability across a large codebase.

Full Stack Developer Intern — September 2023 - April 2024
University of Ottawa — Ottawa, ON

- Redesigned the University of Ottawa's emergency alert website with **C#** and **Angular**, introducing a modern, interactive UI. Used by over **20,000** students daily.
- Optimized existing **C#** REST APIs by reducing **SQL** query execution time for event data lookup by **30 %** and lines of code by **40 %**
- Revamped the event registration and admin webpages by using **Angular** material UI components to improve UI layout and reducing redundant lines of code by almost **50 %**

Data Engineer Intern — May 2023 - August 2023
Trend Micro Inc. — Ottawa, ON

- Designed and implemented a production serverless data pipeline using **AWS Lambda, S3, and Athena**, leveraging **AWS SAM** and **CloudFormation** for infrastructure-as-code and automated deployment.
- Enabled scalable data querying workflows by integrating **Amazon Athena** to query data stored in S3, reducing manual data processing efforts.
- Configured **AWS CloudWatch** and **Splunk** monitoring to track pipeline performance, logs, and failures, enhancing observability and system reliability.

Cloud Developer Intern — September 2022 - December 2022
Trend Micro Inc. — Ottawa, ON

- Developed production serverless data processing pipelines using **AWS Lambda** to analyze client files, images, and code stored in **Amazon S3**, performing vulnerability scans using Anchore APIs and Grype.
- Implemented concurrent processing and robust error handling in **GoLang** using goroutines within Lambda functions, improving reliability and reducing data corruption in high-throughput environments.
- Enabled scalable data querying workflows by integrating **Amazon Athena** to query data stored in S3, reducing manual data processing efforts.

Full Stack Developer Intern — January 2022 - April 2022
Applied Mind Inc. — Ottawa, ON

- Containerized a full-stack application with **Docker** for reliable **production deployments**, and automated event detection by adding events to **PostgreSQL** using **Python** and **GraphQl**, reducing testing time by **10 seconds**.